



VARIATIONS IN K^+ – Na^+ ION EXCHANGE DEPTH IN COMMERCIAL AND EXPERIMENTAL FLOAT GLASS COMPOSITIONS

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ABSTRACT

We report the results of ion-exchange experiments conducted on 17 commercial soda–lime–silicate (SLS) float glass and 8 experimental SLS glass compositions. A significant variation in the depth of K^+ penetration with relatively small changes in composition was observed. The data were fit to a multiple regression model in which the major oxides are the independent variables and depth of K^+ is the dependent variable. The model indicates that increased depth of exchange (increased interdiffusion coefficient) correlates predominantly with increased K_2O and/or Na_2O content of the glass, with a decreased total alkaline earth content and with the ratio of CaO/MgO . © 2000 Elsevier Science Ltd

KEYWORDS: A. glasses, D. diffusion

INTRODUCTION

Chemical strengthening of soda–lime–silicate (SLS) glasses can be achieved through the exchange of K^+ for Na^+ [1]. The relatively larger K^+ ions are “stuffed” into sites occupied by the smaller Na^+ ions, inducing a compressive stress near the glass surface. The most common method of chemical strengthening SLS glass is to place the piece in a molten KNO_3 bath at sub- T_g temperatures. The rate of exchange is controlled by the self-diffusion coefficients of the two alkalis such that the slowest diffusing species, in this case K^+ , governs. The exchange rate is a function of temperature as well as material dependent parameters such as composition, structure, and surface condition of the glass.

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Although chemical strengthening is quite effective, it is presently not in wide commercial use due to the high costs associated with the long treatment times required for a sufficiently deep compression layer. Our primary goal in this study was to determine whether there are significant differences in interdiffusion rates among commercial glass compositions. With this information as well as data from experimental SLS glasses, one can then design new compositions with improved ion-exchange kinetics while maintaining other important properties such as chemical durability.

EXPERIMENTAL

Seventeen commercial float glass samples of differing composition (Table 1) were obtained directly from five manufacturers. Each sample was cut into 5.1×5.1 cm squares and cleaned with alcohol. In addition, four experimental glass compositions produced by O'Connell [2] and four compositions produced for the current work (Table 2; we will refer to them as "DOE glasses") were drawn into rods. All of the samples were immersed in a 450°C high-purity KNO_3 bath for 24 h. This temperature is close to that used in commercial chemical strengthening processes and is below that at which significant stress relaxation occurs for the glasses used here. While 24 h is somewhat longer than the times used for most commercial strengthening processes, the deeper exchange layers associated with longer immersion times are more accurately analyzed.

The depth of exchange for each sample was measured on a scanning electron microscope using energy-dispersive spectroscopy (EDS) in line scan mode. A requirement for an accurate EDS measurement is a sharp 90° edge and, for the commercial glasses, these were obtained by making a line of indentations with a Vickers indenter and spaced at intervals of 1 cm. A drop of water was placed on the indentations and the pieces were broken. Straight edges for the rods were obtained by scoring with a diamond blade and breaking.

Each EDS line scan consisted of six scans with 256 individual channels. Beam conditions were set at 20 kV accelerating voltage and 150 μA emission current. A representative line scan is shown in Figure 1. We measured the depth as the point where the K signal returns to the background level. Two to five replicates were made for each sample. The air sides of the commercial glasses were analyzed, although several analyses of the tin sides were performed. In the present study, there was no significant difference between the depth of the tin and air sides. However, at shorter exchange times, the tin sides usually show significantly shallower exchange depths [3].

RESULTS

The averaged depth of exchange for the float glass samples ranged from 27.7 to 51.0 μm with a standard deviation usually of $\pm 1 \mu\text{m}$ (Table 1). The depth of exchange varies with the square-root of time:

$$\text{depth (cm)} = 2 \sqrt{2D * \text{time (sec)}} \quad (1)$$

where D is the apparent diffusion coefficient. Using this relationship, the calculated D for the slowest float composition is $1.11 \times 10^{-11} \text{ cm}^2\text{-sec}^{-1}$. For the fastest float composition, D is $3.76 \times 10^{-11} \text{ cm}^2\text{-sec}^{-1}$, a 340% increase in the diffusion rate. In terms of time, it would

TABLE 1
Commercial Float Glass Compositions (mol%)^a

Glass	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	Na ₂ O	K ₂ O	Avg. depth (μm) ^b	SD	No. of analyses
1	71.56	0.06	0.04	9.26	5.72	13.15	0.02	29.4	0.2	2
2	71.35	0.97	0.05	9.07	4.98	13.16	0.26	36.0	0.6	4
3	71.29	0.10	0.18	9.50	5.59	13.16	0.04	31.1	1.3	4
4	71.01	0.85	0.04	8.33	5.69	13.28	0.63	39.8	0.5	4
5	71.90	0.07	0.19	9.25	5.63	12.94	0.013	29.7	0.3	3
6	73.39	0.95	0.00	10.95	0.12	14.45	0.006	40.7	2.2	5
7	69.84	0.67	0.04	9.12	6.15	13.80	0.16	30.1	0.3	3
8	71.92	0.83	0.29	8.77	5.21	12.60	0.22	27.7	0.6	3
9	71.87	0.84	0.20	8.81	5.23	12.66	0.22	28.0	0.3	3
10	71.24	0.36	0.04	9.31	5.63	13.13	0.08	33.9	0.6	3
11	71.10	0.07	0.01	9.17	5.86	13.61	0.01	27.6	0.2	3
12	71.47	0.08	0.03	9.08	5.58	13.56	0.04	32.4	0.5	4
13	71.64	0.08	0.05	9.26	5.44	13.32	0.04	30.2	1.6	4
14	71.75	0.33	0.04	9.11	5.36	13.21	0.05	31.7	0.6	3
15	71.69	0.35	0.58	9.07	5.19	13.03	0.08	27.1	0.9	3
16	73.02	1.00	0.01	10.67	0.28	14.80	0.02	42.2	0.9	4
17 ^c	66.97	2.01	0.03	8.49	4.57	17.43	0.48	51.0	0.8	2

^aCompositions were provided by each manufacturer; contact authors for details.

^bAir side; 24-hour treatment at 450°C.

^cNo longer available.

TABLE 2
Experimental Compositions

SiO ₂	67.00	69.75	69.00	70.00
Al ₂ O ₃	1.25	0.25	0.25	0.50
CaO	4.50	5.25	5.75	5.75
MgO	7.50	8.00	7.50	7.75
Na ₂ O	17.00	15.25	14.50	15.50
K ₂ O	2.75	1.50	3.00	0.50
Avg. depth (μm) ^a	89.4	67.4	70.6	57.4
SD	1.3	0.4	0.8	1.0
Number of analyses	3	4	3	5

^a24-hour treatment at 450°C.

take the slower exchanging composition 81 h to reach an exchange depth of 51.0 μm at 450°C.

Our data demonstrate that there is a considerably large range in ion-exchange rates of existing commercial float glass compositions. The depths of exchange for “regular” low alkali float glasses (low K₂O and low Al₂O₃) tend to cluster in the 28–30 μm region while more exotic glasses demonstrated significantly greater depths of exchange. The exotic glasses usually have higher alkali (but not necessarily higher K₂O), high Al₂O₃, and usually lower alkaline earth content.

The exchange depth for the experimental glasses ranges from 56.5 to 89.4 μm with a standard deviation usually of ±1 μm. Relative to the commercial glasses, the experimental glasses are usually higher in K₂O and total alkali, and lower in total alkaline earth content. The experimental glasses of O’Connell [2] only vary in the MgO/CaO ratio.

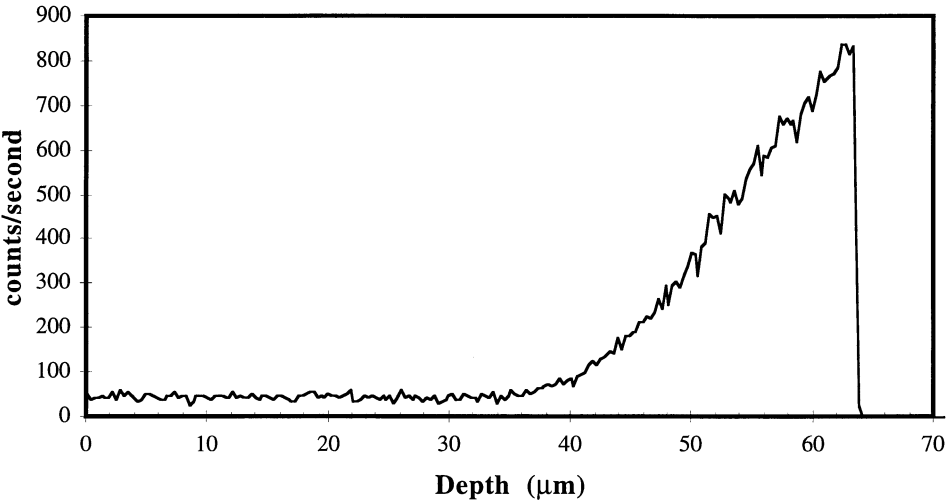


FIG. 1
EDS linescan of sample 8 with K⁺ exchange depth measured at 27.7 μm.

DISCUSSION

With these data, we can attempt to quantify the relationship between composition and diffusion rate (i.e., depth of K^+ exchange in 24 h) and, in turn, use this information to predict the K^+ exchange of a SLS glass. In order to do this, we manipulate the data in two ways: (1) show individual oxide and oxide ratios vs. average exchange depth for each glass composition, and (2) create a linear multiple-regression model that takes into account variations from all of the oxides. The oxide-depth plots enable a visual analysis of the data, but apparent correlations may reflect changes in other oxides or interactions between several oxides. The regression model allows a quantitative examination of the relative significance and role of each oxide in determining the depth of K^+ exchange. In addition, the model produces a linear equation for predicting the ion-exchange rate of an unknown glass.

Figure 2 shows selected oxide and oxide ratios vs. K^+ exchange depth plots. For the regression model, we used JMP statistical software (SAS Institute Inc., Cary, NC), setting the individual oxides as the independent regressors and the depth as the response or dependent variable. The program yields parameters for the y-intercept and the coefficients for each oxide as well as estimates for the fit of the model and the significance of each component. We must make it clear that the compositions used in this study are not part of a statistically rigorous experimental plan and, therefore, are not an ideal data set for a regression model. Nevertheless, we did obtain meaningful results.

Using all seven major oxides, the model had a good fit (R^2 of 0.98), but only the two alkalis were significant (p values < 0.05) and the error for the parameter coefficients was relatively high. When the least significant oxide, Fe_2O_3 , was removed and the model was run again with the remaining six oxides, the model still retained a good fit (R^2 of 0.98), and all of the oxides except CaO were significant in the model (Table 3). However, the work of O'Connell [2] shows that CaO is indeed a significant factor in the exchange rate, so we do not exclude CaO from further discussion. The regression coefficients are a measure of each oxide's relative importance in the depth of exchange (rate of diffusion). The results show that K_2O (16.75) has the greatest effect in increasing the exchange rate, closely followed by Na_2O (13.70). The next lowest coefficients are SiO_2 (10.40) and Al_2O_3 (8.48). The lowest coefficient values are for MgO (7.15) and CaO (3.63).

The intercept and the parameters are used in a predictive equation of

$$K^+ \text{ exchange depth } (\mu m) = y\text{-intercept} + \sum B_i X_i \quad (2)$$

where B_i are the regression coefficients that represent the measure to which the i th oxide affects exchange depth and X_i is the molar percent of each oxide. This equation can be used to predict the K^+ exchange depth of a glass immersed for 24 h in a 450°C KNO_3 bath. The fit of the model is illustrated in Figure 3, which shows a scatter plot of the actual depths vs. predicted depths. Visually, the fit appears to be good, although there is some scatter at lower exchange depths where compositional variations are small. We speculate that the scatter can be attributed to a noncompositional variable such as differences in surface structure.

With the oxide-depth plots and the model results, we can examine the effect of each oxide. Figure 2a shows that the MgO/CaO ratio of the commercial glasses has no apparent effect on the exchange depth. However, the experimental glasses from O'Connell [2], in which only MgO and CaO are varied, clearly indicate an increase in penetration depth with increasing MgO/CaO when the other components are held constant. The effect seems to be most dramatic between 0 and 0.5 MgO/CaO . The higher coefficient for MgO (7.15) relative to

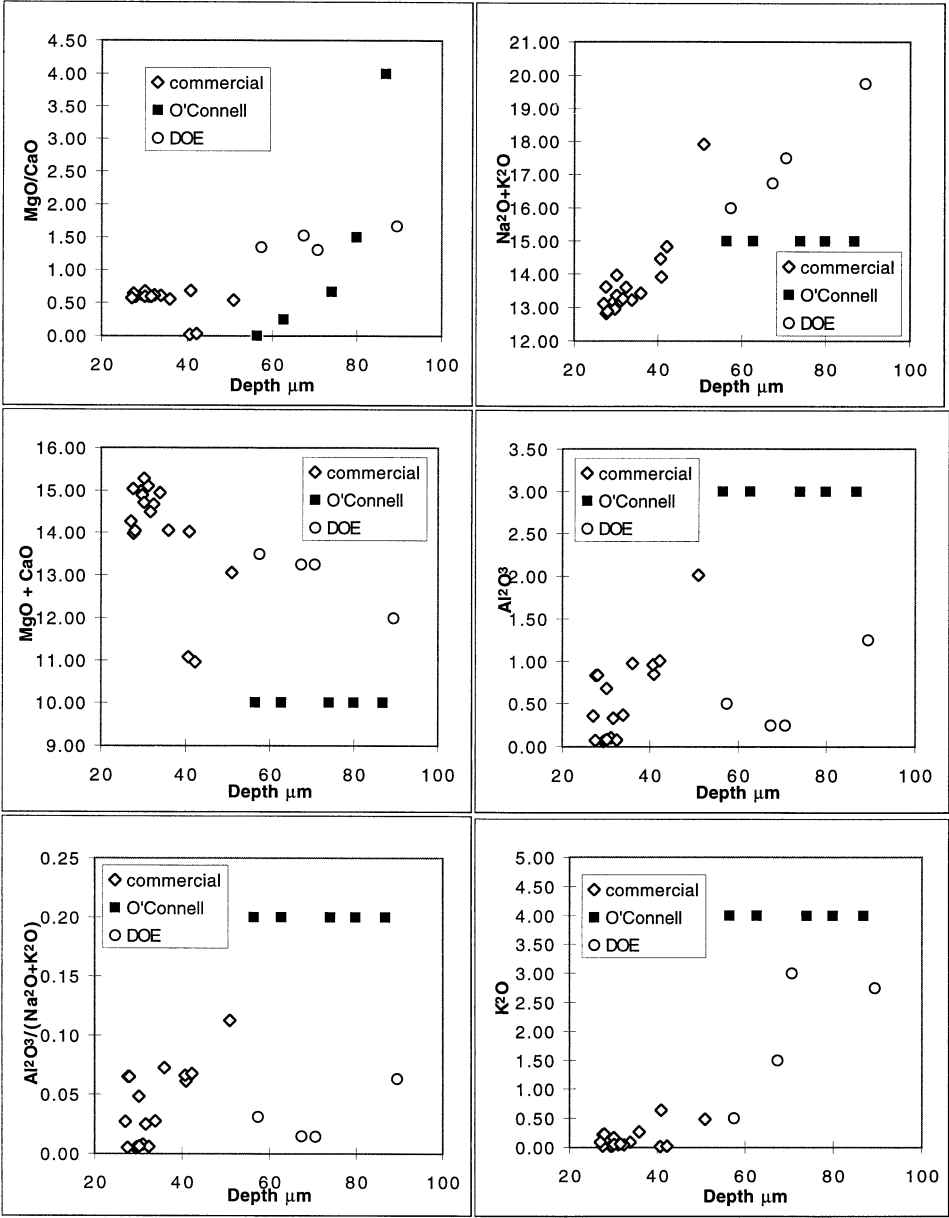


FIG. 2
Selected oxide and oxide ratios vs. K⁺ exchange depth plots.

CaO (3.63) in the regression model also indicate that increased MgO/CaO enhances the exchange rate. O'Connell [2] speculated that the replacement of MgO for CaO results in a more open structure, thus decreasing the activation energy required for diffusion. Figure 2b shows an apparent negative relationship between total alkaline earth content and exchange

TABLE 3
Results of Linear Regression Model

Summary of Fit				
	R ²		0.98	
	Root mean square error		2.27	
	Mean response		42.36	
	Observations		86	
Parameter Estimates				
Term	Estimate	Standard error	t Ratio	Prob. > t
Intercept	−970.06	275.26	−3.52	0.0007
SiO ₂	10.40	2.76	3.76	0.0003
Al ₂ O ₃	8.48	2.88	2.94	0.0043
CaO	3.63	2.87	1.27	0.2095
MgO	7.15	2.83	2.52	0.0136
Na ₂ O	13.70	2.66	5.15	<0.0001
K ₂ O	16.75	2.62	6.39	<0.0001

For a given composition, each individual EDS measurement was used (not the averaged depths).

depth for both the commercial glasses and the DOE glasses. Again, this is consistent with the model that shows MgO and CaO having the lowest regression coefficients.

Previous studies have shown that mixing the alkalis of a SLS glass increases the K⁺–Na⁺ interdiffusion rate [4–6] and our results are consistent with these findings. Figure 2c shows a weak positive correlation between K₂O and exchange depth, although the model indicates that K₂O has the greatest positive effect on diffusion rate. Figure 2d shows a strong positive trend between total alkali content and exchange depth. The mixed-alkali defect model [7,8] suggests that each alkali creates its own specific environment in the glass structure, and so the increased K₂O in the glass produces K “friendly” sites, lowering the activation energy. The positive relationship between total alkali content and exchange rate may be explained by transitional structures [9]. This theory states that some glasses, including SLS glasses, have a structure transitional between isolated alkali regions to interconnected alkali regions. Increasing the alkali content of the glass increases the alkali-rich regions, thereby providing conduits for alkali diffusion.

Figure 2e shows an apparent positive relationship between alumina and exchange depth. However, the commercial glasses with higher Al₂O₃ tend also to have higher alkali content to keep the viscosity curve within the range required for manufacturing. To investigate this possibility, we plotted Al₂O₃/alkali vs. exchange depth (Fig. 2f). The results show that increasing alumina relative to alkali does seem to correlate with increased exchange depth. The model results do not clearly show the role of Al₂O₃, as its coefficient in the model is approximately a median value between the alkalis and the alkaline earths. Previous studies show mixed results as well, with Kim [10] indicating that Al₂O₃ decreased the ion exchange rate, although it does increase the strain point of the glass, reducing the rate of stress relaxation. Mattos [5] found that Al₂O₃ reduced the exchange rate at 450°C, but appeared to increase the exchange rate at higher bath temperatures, particularly for mixed alkali glass compositions.

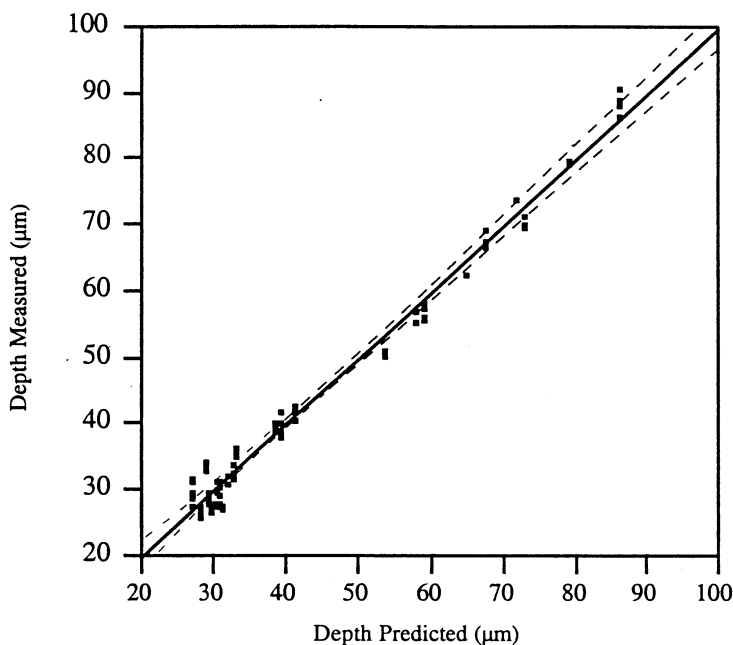


FIG. 3

Predicted depth vs. measured depth for commercial and experimental float glasses exchanged for 24 h at 450°C. Dashed lines represents the 95% confidence limits.

Our results can be used by firms employing chemical strengthening to select a commercial composition capable of being exchanged significantly more rapidly than “normal” float glass. For instance, use of compositions 6 and 16 rather than 5 or 8 for exchange would permit the processor to reduce the time required to reach a particular depth of exchange. For example, eq. 1 indicates that substitution of composition 16 by composition 8 would reduce the exchange time from 19.6 to 8.4 h for a 25 μm exchanged layer.

The data are also useful for designing new compositions with increased exchange rates. However, one must recognize that several other properties (chemical durability, viscosity) must be either unaffected or improved in the new composition, and manufacturing considerations need to be taken into account (raw materials cost, refractory corrosion, melting and firing temperatures). Furthermore, there is a complex dependence of diffusion rates on composition and exchange temperature. As noted, alumina may either increase or decrease interdiffusion rates, depending on composition and on the temperature of exchange. Clearly, if more exact predictions of ion-exchange rates are to be generated, a substantial number of additional experimental melts will be required. This work is currently underway (DOE Contract DE-FC02-98CH10920).

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